Fivetran

TECHNICAL OVERVIEW

Fivetran HVR: High-volume, real-time data replication

High-volume data integration simplified

The Fivetran data movement platform can integrate all of your SaaS, file, event and database systems. For enterprise log-based database replication, Fivetran HVR simplifies high-volume, real-time data movement. Our simple, yet powerful data integration and synchronization solution reduces infrastructure costs while allowing you to create new and enhanced services driven by data stored on-premises and in the cloud.

With Fivetran HVR, you can replace your batch processes with a variety of near real-time data delivery and integration scenarios as well as enable consolidated, near real-time analytics that improve your business insights. This white-paper describes the technologies and capabilities available, covering:

- Use cases and topologies
- Secure and efficient data transfer
- Flexible architectures
- Integrated, all-in-one capabilities
- Broad platform support

During the first few decades of IT automation organizations implemented database-based applications on-premises. Such implementations are still very common, despite the popularity of cloud-based deployments. But individual applications are data silos, and organizations recognize the value of consolidated access to data for analytics. With the growth in data volumes, building consolidated, high performance and scalable data analytics environments has become complex and very costly.

To address these challenges, enterprises are modernizing their infrastructure and adding cloud data warehouses and data lakes to their data platforms to keep up with growing data demands. This is creating new challenges for your data to move between systems - seamlessly, automatically and securely.

Use cases and topologies

Fivetran HVR is a data replication and validation solution that works across complex, heterogeneous environments. It works with major on-premises and cloud solutions, data lake solutions and data formats. This flexibility makes it ideal for a wide range of data analytics, operational and transactional use cases. We support these use cases through the following topologies:



Uni-directional

Move data in one direction, from source to target. This is the most common topology. Organizations use this topology to feed a data warehouse or data lake, and for data migrations projects.



Consolidation

A consolidation topology takes data from multiple identical data stores and merges it into a central data store, data warehouse or data lake. Fivetran HVR gives you the option to add columns and set metadata values to enable uniqueness across multiple identical sources.



Broadcast

To deliver data from production systems into multiple destinations, for example a data lake and a data warehouse, or simply to distribute data, the broadcast topology captures data from a single source and delivers it to multiple targets. Fivetran HVR transfers the same data everywhere or you can use filtering to distribute different data from the source database to different targets.



Cascading environment

In this use case, Fivetran HVR allows you to take data from one source, put it in a centralized operational data store, and then broadcast it to several downstream data warehouses or data marts.

Secure and efficient data transfer

As your data moves across your infrastructure, you need to ensure your data is secure and protected. Fivetran HVR offers robust security from source to target, in transit and at rest.

HVR uses TLS 1.3 to secure data in transit between a high-volume agent and the HVR hub. Data encryption at-rest secures data stored on disk. In a distributed setup, the firewall only needs to be opened on a single port in a single direction for the HVR hub server to reach the high-volume agent.

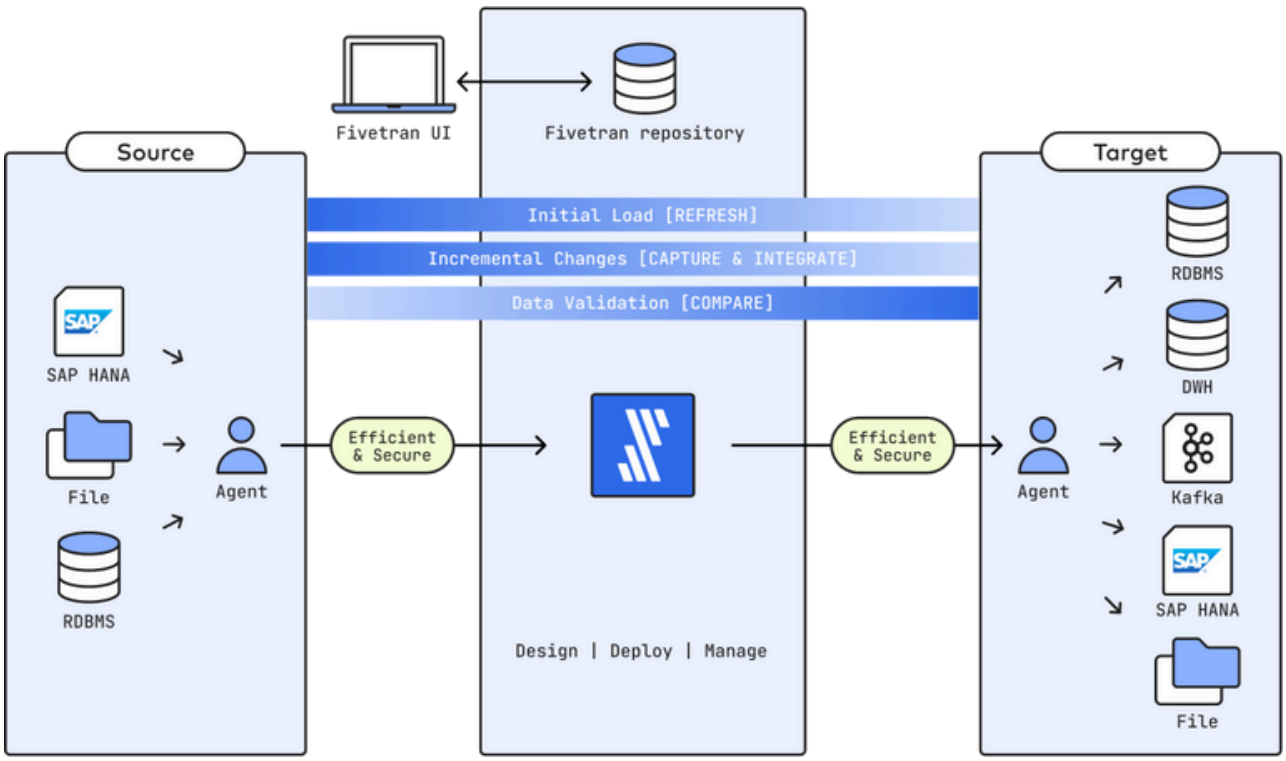
Fivetran HVR improves the efficiency of data transport by capturing only relevant changes from the source and then compressing the data as soon as it is captured. Data always moves compressed between the high-volume agent and HVR hub, and is only decompressed just before delivery into the target. To deliver data into a target, Fivetran HVR uses high-speed native technologies, including clustered file systems, staging tables and other technology-appropriate options for maximum performance.

Most data warehouses are not architected to handle continuous updates from busy transaction processing systems. Fivetran HVR's micro-batch (burst) approach provides an optimized way to transfer changes in near real-time. The burst method first stages the data. It then generates set-based SQL statements to merge changes into the target.

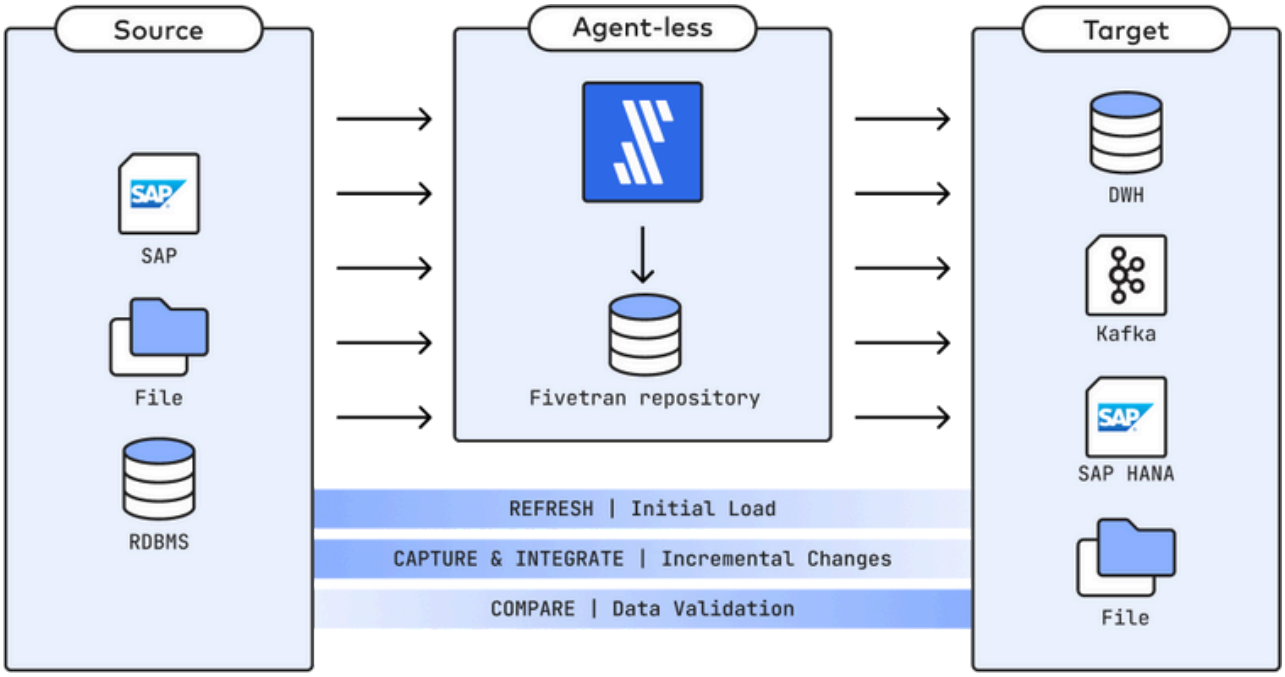
A flexible architecture

Fivetran's technology is designed to fit into your architecture. With Fivetran HVR, we recommend a distributed setup with a high-volume agent installation and a central HVR hub for three reasons:

1. **Performance**: the high-volume agent identifies relevant changes on the source with low latency log-based Change Data Capture (CDC) before data is passed to the destination. All data transfer between the high-volume agent and HVR is densely compressed and uses optimum network communication.
2. **Scalability**: individual agents perform part of the work, offloading resource-intensive processing from the hub.
3. **Security**: data transfer between the high-volume agent and HVR is always encrypted using TLS 1.3.



However, Fivetran HVR also supports agentless connections to sources and destinations that are an easier-to-configure alternative for environments requiring less performance or scalability.



The HVR hub server manages all configurations, controls all processing and directs data where it needs to go. In addition, the hub collects statistics and maintains metadata about the Fivetran ecosystem.

The hub server is managed through a browser-based interface. It is used to define the replication, schedule full and partial data loads, start/stop capture and apply jobs, as well as perform data validation and repair. The hub server can be deployed anywhere in the infrastructure—on the source, target or its own dedicated server. For most use cases with more than two endpoints connected, the hub server is deployed on a separate middle tier environment.

A complete data movement platform: An all-in-one, integrated solution

Fivetran HVR offers everything you need to set up and manage near real-time data integration. You can integrate dozens of data storage platforms, services and formats using one small download. The comprehensive data replication and validation solution includes:

- Centralized management/administrative console to configure, manage and maintain data replication pipelines.
- Target table creation and initial data load with automated table and data type mapping, including support for automatic propagation of data definition changes (DDL) in heterogeneous configurations.
- Real-time, high-volume, log-based change data capture (CDC) on a multitude of database technologies.
- Cross platform data validation and repair between databases.
- Rich monitoring and reporting.

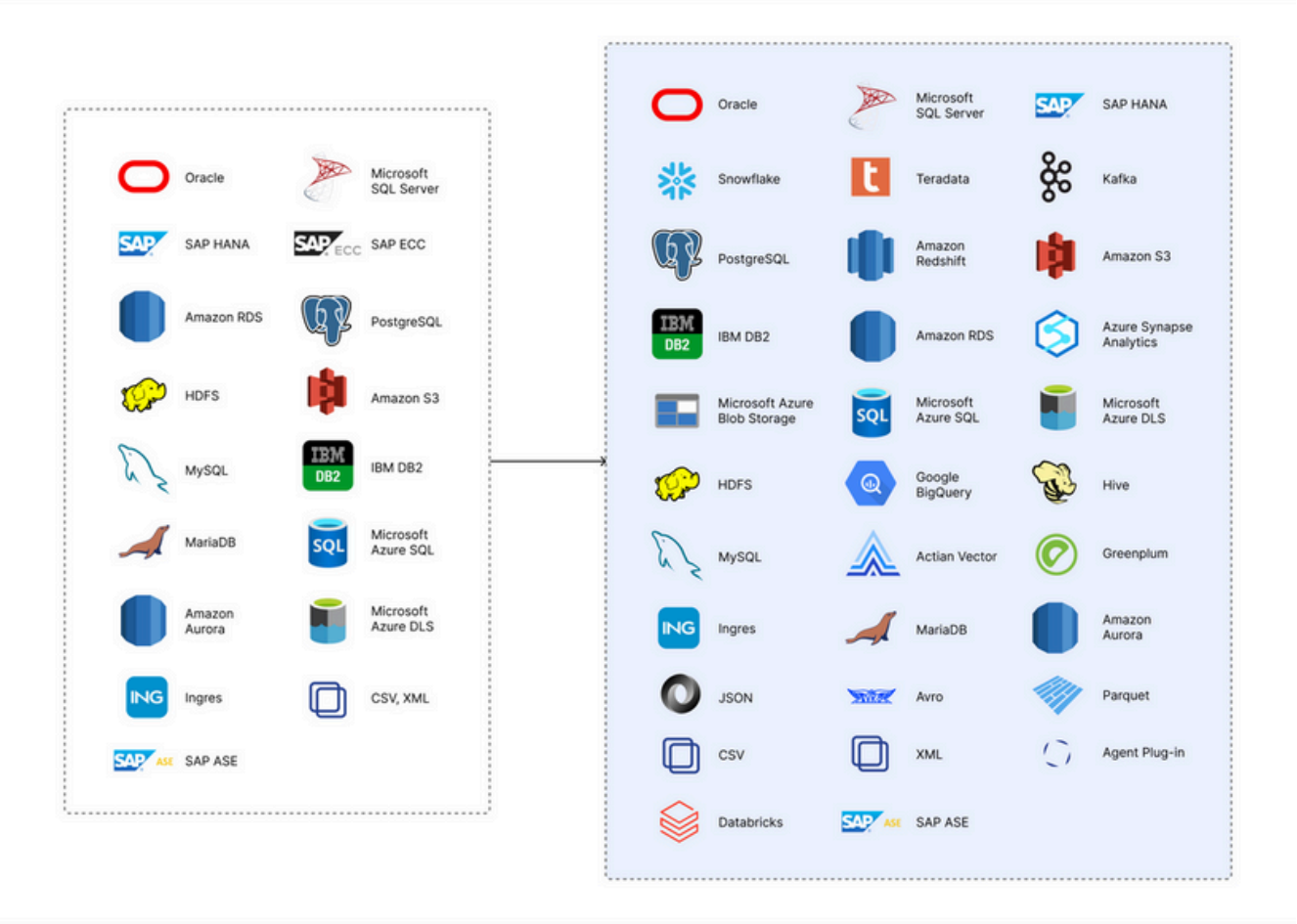
Broad platform support

Bridging the gap between traditional relational database applications and modern cloud-based data systems, Fivetran HVR supports an extensive range of sources and targets, shown in the diagram below. HVR supports log-based Change Data Capture on all supported source database technologies, including SAP HANA.

Due to the complexity of SAP ERP, the HVR technology provides unique capabilities to streamline low-impact, log-based CDC from the enterprise application suite:

- Table selection for SAP is through modules, based on table descriptions and with the ability to see related tables based on the SAP dictionary.
- Support for decoding of cluster, pool and long text tables in long-time SAP ECC deployments that have not migrated to SAP HANA.
- Full support for customer Z tables.
- The ability to access the database through Netweaver instead of a direct database connection, to allow for database runtime licenses.

Fivetran HVR supports commonly-used cloud-based platforms including Snowflake, Google BigQuery and Databricks, as well as numerous other technologies. An Agent Plugin script or program can be used to deliver changes into a destination that is not natively supported by Fivetran HVR. Plugin examples are available on Github.



Getting started

For flexible, high-volume and near real-time data integration between databases and cloud-based environments, look no further than Fivetran.

Reach out to sales@fivetran.com to learn more about our data movement platform.